

# From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

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**TEAM OUTLINE.** Section content is stated at outline level only, with status and owner per section. The worked detail for the classical models (Levels 1 and 2) lives in `classicalmodelling_notes.tex`, including the annotated reference checklist and the margin-diagnostic derivation. Status key: [OUTLINE] skeleton only, [DRAFTED] argument settled in notes, [DATA] waiting on measurement, [BLOCKED] waiting on a named item.

**Abstract**—Effective low-level feedback control of small under-actuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide both actionable design insight and predictive accuracy. Tow-tank testing is impractical at this scale; parameters must be identified solely from limited field trials. Existing system identification methods span a wide range of model complexity but offer little guidance on how much fidelity is enough, because sufficiency is only meaningful relative to a stated purpose. Absent that specification, the default is a bias toward more fidelity, since the benefit of an added term is readily quantified while its costs in identifiability, experimental burden and obscured design insight are not.

We propose a methodology built around a deliberate spectrum of three model levels, each identified from the same experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions, Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from step-response field experiments demonstrate the approach. First-order SISO models achieve  $R^2=0.93$  (surge,  $\tau=1.95$  s) and  $R^2=0.83$  (yaw rate,  $\tau=0.47$  s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness, a key coupling for underactuated control design, is not resolvable from a single short trial, motivating targeted data collection. A margin-based diagnostic makes the sufficiency question computable, reporting the forward speed at which the first-order model stops changing the control design. The NARX model matches one-step prediction accuracy ( $R^2=0.93$ ) but diverges in closed-loop simulation, demonstrating that physical structure provides essential regularization under limited data. A larger dataset spanning multiple operating regimes is being collected to evaluate the full methodology.

**Index Terms**—uncrewed surface vehicle, system identification, model spectrum, control-oriented modeling, robustness margin,

sim-to-real transfer, nonlinear dynamics, underactuated vessels

## STATUS BOARD

| Section            | Status  | Own    | Next action            |
|--------------------|---------|--------|------------------------|
| I. Introduction    | OUTLINE | BB     | Sufficiency placement  |
| II. Related Work   | DRAFTED | BB     | Verify Woo 2018 (CV)   |
| III. Methodology   | OUTLINE | BB, CV | Compute Diagnostic 4   |
| IV. Platform, Data | DATA    | TBD    | Vehicle specifications |
| V. Results         | BLOCKED | BB, CV | Blocked on III, IV     |
| VI. Discussion     | OUTLINE | TBD    | Follows V              |
| VII. Conclusion    | OUTLINE | BB     | Follows VI             |

**Action (team):** Level 1 and 2, the classical models, are BB. Level 3, the black-box NARX model, is CV. Add CV's contact address before submission.

## I. INTRODUCTION

[OUTLINE] [BB]

- Motivate: small USVs under 7 m with a single thruster and rudder; low-level speed and heading control is a prerequisite for higher-level autonomy.
- The tension: “how much fidelity is enough” has no answer until “enough for what” is specified. Absent that, practice defaults to more fidelity, because the benefit of an added term is measurable and its cost is not.
- Framing: Box, all models are wrong but some are useful [1]; a spectrum is collectively more useful than any one complex model.
- The decision ladder D1–D5: what a control-oriented model can support (architecture choice, end-to-end verification, relative comparison, tuning rehearsal) and what it cannot (gains that transfer without field tuning).
- Scope: single thruster plus rudder, under 7 m, field-identified parameters. Distinct from differential-thrust craft and from vessels with tow-tank access.
- Contributions: the three-level spectrum; cross-model diagnostics; a margin-based sufficiency test; a targeted data-collection framework.

**Action (BB):** Decide whether the sufficiency argument stays inside the introduction or becomes a short standalone section before Sec. III. It currently runs long enough to unbalance the introduction.

*See notes: Introduction, including the four concrete costs of excess fidelity, the two independent bounds on useful fidelity and the Box versus Skelton distinction.*

## II. CLOSELY RELATED WORK

[DRAFTED] [BB]

- Response models as the minimal control-oriented description: Nomoto’s first-order steering model [2], its place in the standard formulation [3], and the identification tradition founded by Åström and Källström [4].
- Physics-based maneuvering models and why their coefficients resist identification from field data [5]–[8].
- Control-oriented identification of small craft, led by the nearest antecedent, Sonnenburg and Woolsey [9], with [10]–[13].
- Data-driven and grey-box alternatives [14]–[16], and the one-step versus closed-loop evaluation gap.
- Positioning, three gaps: prior work selects one model rather than retaining the set; data-driven marine models are evaluated on one-step prediction only; the small rudder-steered vessel is underrepresented.

**Action (CV):** Acquire Woo *et al.* 2018 and verify that they report one-step prediction only. Gap 2 rests on this and it is the foil for the Level-3 result. Do not submit without checking.

**Action (BB):** Confirm the Skelton framing carries the “appropriate, not small” criterion as quoted, once the paper is in hand.

*See notes: Closely Related Work, an annotated checklist giving for each reference the locator, why it is cited and what remains unresolved.*

## III. MODEL SPECTRUM METHODOLOGY

[OUTLINE] [BB]

- Level 1: first-order linear SISO transfer functions for surge and yaw, identified as discrete ARX(1,1) by ordinary least squares.
- Level 2: nonlinear state-space with quadratic drag, identified in two phases so that drag is estimated from coasting and thrust gain from powered segments.
- Level 3: NARX multilayer perceptron, trained on a chronological split, with one-step prediction and closed-loop simulation reported separately.
- Cross-model diagnostics, four in total: speed-dependent rudder gain; drag nonlinearity; rudder-induced surge drag; and whether the added fidelity falls inside the robustness margin of the design.

**Action (BB):** Compute Diagnostic 4 on the existing dataset. Minimum viable form is the gain ratio against the Level-1 gain margin plus the crossover speed for surge.

**Action (CV):** Level 3 NARX: confirm the architecture, lag count and chronological split stated here match the current notebook, and settle whether the black-box level is framed as a competitor or as an unregularized reference point.

**Action (CV):** Level 3 lives on a separate branch; Levels 1 and 2 are on traditional-models. Coordinate the merge so Sec. III reads as one piece.

*See notes: Model Spectrum Methodology, and the four-step derivation of Diagnostic 4 with the Level-2 linearization.*

## IV. EXPERIMENTAL PLATFORM AND DATA COLLECTION

[DATA] [TBD]

- Vehicle description, autopilot and sensing. Inputs are normalized RC throttle and rudder; outputs are GPS speed and IMU yaw rate.
- Data cleaning: multi-rate streams aligned to a 10 Hz grid, one gap handled by segmentation.
- Current dataset: a single step-response trial, 130 s working window, throttle 0–39%, speed 0–2.9 m/s. Intentionally minimal, a proof of concept.
- Planned dataset: multiple trials across the throttle range, dedicated rudder sweeps at fixed speeds, and maneuvers spanning the operating regimes.

**Action (TBD):** Fill the vehicle specifications: make, model, length, mass, propulsion and the speed range including any regime transitions.

**Action (TBD):** Plan the rudder-sweep trials. These are what resolve the speed-dependent rudder gain that Diagnostic 1 could not settle.

## V. RESULTS

[BLOCKED] [TBD]

- Level 1: identified time constants and gains, simulation fit, and a measured against simulated figure.
- Level 2: identified parameters, coasting against joint fit for the drag term, and fit comparison against Level 1.
- Level 3: one-step against closed-loop simulation fit, and the divergence figure for surge.
- Cross-model diagnostic summary: four diagnostics, findings and the implication each carries for model and control design.

**Action (CV):** Level 3 results: one-step against closed-loop simulation fit, and the surge divergence figure. This pair is the paper’s central empirical claim.

**Action (BB, CV):** Produce the combined figure showing measured data against all three levels on one set of axes.

**Action (TBD):** Blocked on Sec. III Diagnostic 4 and on the vehicle specifications.

## VI. DISCUSSION

[OUTLINE] [TBD]

- What the spectrum reveals for this vehicle class: weak surge-yaw coupling at the speeds tested, significant quadratic drag, and a speed-dependent rudder gain that the current data can neither confirm nor deny.
- Implications for compensator architecture, level by level.

- Return to D1–D5: none of the three levels reaches D5, and saying so plainly is the honest position. The defensible claim is D1 through D4.
- Limitations of the current dataset and what the full study adds.

## VII. CONCLUSION

[OUTLINE] [BB]

- The spectrum collectively provides more control design guidance than any single model; simple models are a persistent design tool rather than a stepping stone.
- What the full dataset will allow.
- Future work: margin-based sufficiency as a formal criterion across the spectrum, replacing expert judgment with a computable bound.
- Release the dataset and analysis notebooks.

## OPEN DECISIONS FOR THE TEAM

- Page budget. The venue allows a short paper; the material currently exceeds it. Candidate cuts are recorded per reference in the notes.
- Whether the economics argument, that required fidelity is set by the ratio of hardware-test cost to simulation-test cost, is supportable here or should be flagged as future work.
- Whether Diagnostic 4 appears in full or in its minimum viable form, with the full treatment deferred.
- Whether Sec. I keeps the sufficiency argument or yields it to a standalone section.

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